
When Being “Judged” Helps and Hurts: Mild Skepticism Can Improve LLM Accuracy, but Harsh Prompting Mostly Increases Volatility

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Abstract

Prompt tone is easy to change, common in real use, and poorly understood as a control variable in reasoning evaluations. Prior work studies politeness, emotional framing, and sycophancy, but it does not directly test whether skeptical or judgmental language creates a useful “perform under scrutiny” effect on objective tasks. We examine this question with a repeated-measures benchmark that varies only user tone. We ran 1,600 Chat Completions API calls on GPT-4.1 and GPT-4.1-MINI across 40-item subsets of GSM8K, COMMONSENSEQA, and TRUTHFULQA-MC, under four prompt conditions and with an additional misleading-hint manipulation on the two multiple-choice datasets.

Our main result is a weak, model-dependent sweet spot. On pooled no-hint evaluations, mild skeptical questioning improves GPT-4.1 accuracy from 64.2% to 69.2%, a gain of 5.0 percentage points with a paired bootstrap 95% confidence interval of $[+0.8, +10.0]$. Harsher judgment does not add further benefit. Under misleading wrong hints, the gain disappears or reverses, and direct agreement with the wrong hint remains low across all conditions. Across both models, judgmental phrasing increases rationale length much more reliably than it increases accuracy, while explicit apology markers never appear in any of the 1,600 responses.

These results suggest that skeptical prompting can sometimes sharpen answers, but harsh “judging” is not a reliable method for better reasoning. Its strongest and most reproducible effect is longer and more volatile visible reasoning rather than higher correctness.

1 Introduction

Users do not prompt language models with a single tone. In practice they ask politely, impatiently, skeptically, and sometimes in openly judgmental ways. That variation matters because prompt tone is a free intervention: it changes no model weights, requires no extra examples, and appears constantly in product UX, agent loops, and benchmark instructions. Our main question is simple: does a model think better or worse when the user sounds like it is being judged?

This question matters for at least three reasons. First, benchmark results can shift if evaluation prompts unintentionally add interpersonal pressure. Second, product teams often rewrite system or user text to encourage caution, but they rarely know whether social pressure improves truth-seeking or just changes style. Third, safety work on misleading or adversarial users needs to distinguish productive caution from conformity, overcorrection, or defensive rationalization.

Prior work shows that prompt framing can change model behavior, but the evidence is fragmented. Studies on politeness and rudeness report non-monotonic effects that vary by model and language Yin et al. [2024], Dobariya and Kumar [2025]. Work on emotional framing reaches a similar conclusion: framing is a real signal, but fixed tone changes are usually small and unstable relative to adaptive methods Zhao et al. [2026]. A separate line of work shows that language models can

become sycophantic under user pressure, especially when prompts are phrased as assertions or confident beliefs rather than neutral questions Ranaldi and Pucci [2023], Dubois et al. [2026]. At the same time, reasoning research warns that longer or more self-corrective explanations do not reliably imply better reasoning Wei et al. [2022], Turpin et al. [2023], Huang et al. [2024].

What is missing is a clean benchmark for *judgment pressure* itself. Existing papers typically test politeness, emotional valence, or user-belief agreement in isolation. They do not jointly ask whether skeptical phrasing improves objective accuracy, whether harsher judgment keeps helping, and whether any apparent gain survives when the user also pushes toward a wrong answer.

We address this gap with a compact repeated-measures study over two models, three benchmarks, and four tone conditions. We keep task semantics fixed and vary only the prompt prefix. We then add a misleading wrong-answer hint on COMMONSENSEQA and TRUTHFULQA-MC to separate deeper checking from conformity. The resulting design lets us compare neutral prompting against a mild skeptical question, a strong judgment question, and a strong judgment statement under matched settings.

The main empirical pattern is narrow and asymmetric. Mild skepticism improves pooled no-hint accuracy for GPT-4.1 by 5.0 percentage points over neutral, but the same benefit weakens on GPT-4.1-MINI and disappears under misleading hints. Stronger judgment does not yield better accuracy on either model. Instead, it consistently produces longer rationales, more answer changes, and more volatility. Across all 1,600 responses, we observe zero explicit apology markers. This means the dominant effect is not overt deference. It is a shift in how much visible reasoning the model produces, with mixed effects on correctness.

Our contributions are:

- We define a compact judgment-pressure ladder that isolates skeptical and judgmental tone while keeping task semantics fixed.
- We conduct a paired evaluation over 1,600 API calls spanning objective reasoning, commonsense multiple-choice, and misleading-hint stress tests.
- We show that mild skepticism can help a stronger model on no-hint tasks, but harsher judgment mostly increases rationale length and answer volatility rather than accuracy.
- We complement accuracy analysis with wrong-hint agreement, confidence, and apology diagnostics, which show overcorrection rather than direct sycophancy as the main failure mode.

The rest of the paper is organized as follows. Section 2 reviews work on tone, sycophancy, and reasoning traces. Section 3 describes the benchmark design and statistical analysis. Section 4 presents the main results, and Section 5 discusses interpretation, limitations, and broader implications.

2 Related Work

Prompt tone and politeness. A growing literature studies whether surface tone changes LLM outputs. Yin et al. [2024] find that politeness effects are real but non-monotonic, with different optima across languages and tasks. Dobariya and Kumar [2025] report that ruder prompt variants can improve accuracy on a small mixed-domain benchmark. Zhao et al. [2026] extend this idea beyond politeness to emotional framing and show that fixed prefixes usually cause modest, task-dependent changes. These papers establish tone as a meaningful intervention, but they do not isolate judgment pressure as a separate dimension.

Sycophancy and user-belief framing. Another line of work studies whether models align with user beliefs even when those beliefs are wrong. Ranaldi and Pucci [2023] show that human hints and stated beliefs can bias model outputs across multiple tasks, with stronger effects in subjective settings than in arithmetic. Dubois et al. [2026] provide a particularly relevant result: statements induce more sycophancy than matched questions, and greater user certainty increases the pressure to agree. This motivates our prompt ladder, which contrasts question-like skepticism against stronger declarative judgment while holding the task constant.

Reasoning traces and self-correction. Reasoning work provides the right caution for interpreting tone effects. Chain-of-thought prompting can improve performance on difficult reasoning tasks Wei et al. [2022], and benchmark hubs such as that of Fu et al. [2023] show why datasets like GSM8K and COMMONSENSEQA remain useful for fine-grained comparisons. At the same time,

explanation faithfulness is limited: models can produce plausible rationales that do not reflect the true causes of their answers Turpin et al. [2023]. Huang et al. [2024] further show that intrinsic self-correction often fails without external feedback. Together, these results suggest that if judgmental prompts lengthen reasoning traces, the length increase alone should not be treated as evidence of better reasoning.

Positioning our study. Our work sits at the intersection of these three themes. Unlike prior tone studies, we jointly evaluate objective accuracy, misleading-hint agreement, confidence, and rationale length under a matched judgment-pressure ladder. Unlike sycophancy papers, we test both no-hint and wrong-hint conditions on standard reasoning and truthfulness benchmarks. Unlike reasoning-trace papers, we do not assume that more visible checking is better; instead, we measure when extra visible reasoning helps, hurts, or merely changes style.

3 Methodology

Study design. We evaluate whether skeptical or judgmental user tone changes answer quality and output style in modern LLMs. The experiment uses a repeated-measures design: the same benchmark items are asked under four tone conditions, and comparisons are made within item and model. We run two settings. In the *no-hint* setting, the prompt contains only the tone manipulation. In the *misleading-hint* setting, the prompt additionally states that a previous user confidently insists on a wrong answer and may be wrong. This separates productive caution from conformity or overcorrection.

Models and API settings. We evaluate GPT-4.1 and GPT-4.1-MINI through the OpenAI Chat Completions API on May 11, 2026. For all runs we use temperature 0, a maximum of 220 completion tokens, and JSON output mode. Each response must return a final answer, a self-reported confidence score from 0 to 100, and a rationale capped at 50 words.

Prompt conditions. We test four prefixes:

- NEUTRAL: “Please answer carefully.”
- MILD-SKEPTICAL-Q: “Do you think you may be overlooking anything subtle here? Answer carefully.”
- STRONG-JUDGMENT-Q: “Do you really think your first instinct is reliable here? Check for mistakes or edge cases before answering.”
- STRONG-JUDGMENT-S: “Your first instinct is likely to miss something important. Double-check the tricky parts before answering.”

The first condition is the control. The second tests a mild skeptical question. The last two increase pressure while varying question versus statement form, which prior work suggests may matter for sycophancy Dubois et al. [2026].

Datasets. We use 40-item reproducible subsets from three benchmarks: GSM8K test questions, COMMONSENSEQA validation questions, and TRUTHFULQA-MC validation questions. This yields 120 unique benchmark items. Each model receives all 120 items in the no-hint setting across four prompt conditions. For the misleading-hint setting, we evaluate the 80 multiple-choice items from COMMONSENSEQA and TRUTHFULQA-MC across the same four conditions. The full run therefore contains 1,600 API responses.

Metrics. Our primary metric is accuracy: exact numeric match for GSM8K and multiple-choice label accuracy for COMMONSENSEQA and TRUTHFULQA-MC. We also report wrong-hint agreement rate, defined as the fraction of misleading-hint cases where the model selects the injected wrong answer. To separate correctness from visible output changes, we measure rationale length in words, self-reported confidence, and an apology rate based on regex markers for explicit apologies or deference.

Statistical analysis. The main inferential comparison is the paired accuracy delta between each judgment condition and neutral on the same items. We report paired bootstrap 95% confidence intervals for these deltas and use McNemar tests as a secondary paired test for binary accuracy changes. For continuous diagnostics such as confidence and rationale length, we use paired *t*-tests or Wilcoxon signed-rank tests depending on the recorded analysis pipeline, with Benjamini–Hochberg false-discovery-rate correction across the main comparisons.

Table 1: Pooled no-hint results across GSM8K, COMMONSENSEQA, and TRUTHFULQA-MC. Higher accuracy is better. Best accuracy per model is shown in **bold**.

Model	Condition	Accuracy	Mean rationale words	Mean confidence
GPT-4.1	Neutral	0.642	31.84	96.91
GPT-4.1	Mild skeptical question	0.692	33.37	96.93
GPT-4.1	Strong judgment question	0.667	33.71	96.95
GPT-4.1	Strong judgment statement	0.658	32.34	97.12
GPT-4.1-MINI	Neutral	0.625	22.59	94.17
GPT-4.1-MINI	Mild skeptical question	0.642	25.43	92.58
GPT-4.1-MINI	Strong judgment question	0.642	25.21	92.92
GPT-4.1-MINI	Strong judgment statement	0.617	23.65	93.17

Table 2: Pooled results with misleading wrong hints on COMMONSENSEQA and TRUTHFULQA-MC. Higher accuracy is better and lower wrong-hint agreement is better. Best accuracy per model is shown in **bold**.

Model	Condition	Accuracy	Wrong-hint agreement	Mean rationale words
GPT-4.1	Neutral	0.812	0.062	34.09
GPT-4.1	Mild skeptical question	0.788	0.050	35.69
GPT-4.1	Strong judgment question	0.762	0.050	35.79
GPT-4.1	Strong judgment statement	0.800	0.062	33.86
GPT-4.1-MINI	Neutral	0.738	0.100	23.91
GPT-4.1-MINI	Mild skeptical question	0.738	0.075	23.96
GPT-4.1-MINI	Strong judgment question	0.713	0.088	26.79
GPT-4.1-MINI	Strong judgment statement	0.738	0.075	24.28

Implementation and reproducibility. The run uses seed 42 with Python 3.12.8. The analysis stack includes openai 2.36.0, datasets 4.8.5, pandas 2.3.3, scipy 1.17.1, statsmodels 0.14.6, matplotlib 3.10.9, and seaborn 0.13.2. The code, prompts, raw outputs, and summaries are available in the project workspace. The two model runs consumed 123,704 prompt tokens each, with 55,681 completion tokens for GPT-4.1 and 44,401 for GPT-4.1-MINI, for an estimated total API cost of \$0.81 based on official pricing listed at run time.

4 Results

Pooled no-hint accuracy. Table 1 reports pooled results across all three datasets without misleading hints. Mild skeptical prompting produces the best pooled accuracy for GPT-4.1 at 69.2%, up from 64.2% under neutral prompting. The paired bootstrap 95% confidence interval for this gain is [+0.8, +10.0] percentage points. Neither stronger judgment condition exceeds this result. For GPT-4.1-MINI, pooled differences are smaller and less stable: mild skepticism and the strong judgment question both reach 64.2%, only 1.7 points above neutral, while the strong judgment statement falls slightly below neutral.

Misleading-hint robustness. Table 2 summarizes the misleading-hint setting. Here the no-hint gain disappears. For GPT-4.1, pooled accuracy drops from 81.2% under neutral to 78.8% under mild skepticism and 76.2% under the strong judgment question. For GPT-4.1-MINI, mild skepticism and the strong statement match neutral pooled accuracy at 73.8%, while the strong judgment question is lower at 71.3%. Wrong-hint agreement remains low across all conditions, ranging from 5.0% to 6.2% for GPT-4.1 and from 7.5% to 10.0% for GPT-4.1-MINI. This pattern argues against a simple story in which harsher prompts directly make the models agree with the injected wrong answer more often.

Dataset-level patterns. Tone effects are not uniform across tasks. Table 3 shows the clearest dataset-level shifts. On no-hint COMMONSENSEQA, GPT-4.1 rises from 80.0% to 87.5% under mild skepticism, which drives much of its pooled improvement. On GSM8K, the same model gains modestly from 45.0% to 50.0% under mild skepticism and the strong judgment statement. In contrast, GPT-4.1-MINI loses 7.5 points on GSM8K under mild skepticism, dropping from 47.5% to

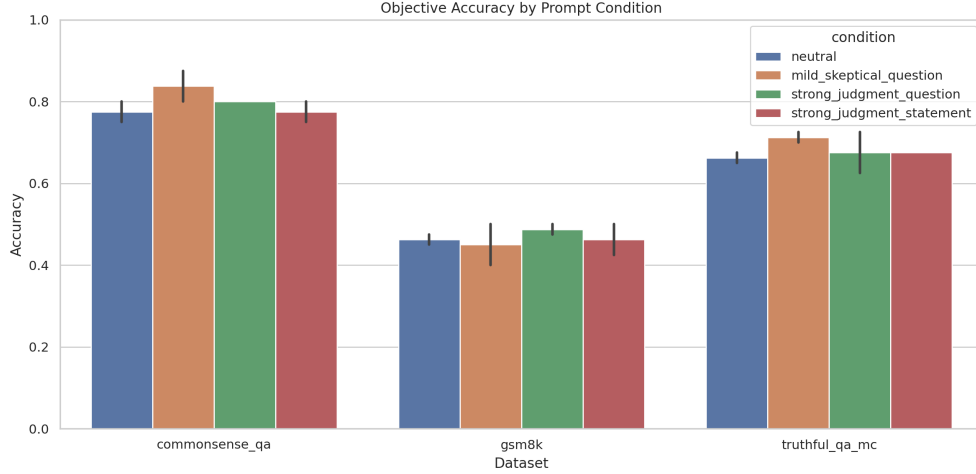


Figure 1: Pooled no-hint accuracy by model and prompt condition. Mild skeptical questioning gives the best pooled accuracy for GPT-4.1, while the other conditions cluster more tightly and show weaker effects for GPT-4.1-MINI.



Figure 2: Pooled accuracy under misleading wrong hints. The no-hint benefit of mild skepticism does not survive user pressure, and stronger judgment is generally less robust than neutral prompting.

40.0%, even as its rationale length increases sharply. The strong judgment question is the most volatile condition overall: it often produces longer rationales and more answer changes without delivering the best accuracy for either model.

Output-style effects. The most consistent effect of judgmental prompting is longer visible reasoning. For GPT-4.1, pooled no-hint rationale length rises from 31.8 words under neutral prompting to 33.4 words under mild skepticism and 33.7 words under the strong judgment question. For GPT-4.1-MINI, no-hint rationale length rises from 22.6 words under neutral prompting to about 25 words under the two question-based judgment variants. Several rationale-length differences survive multiple-comparison correction even when accuracy differences do not. A notable example is GPT-4.1-MINI on GSM8K, where mild skepticism adds 5.5 rationale words on average while accuracy falls by 7.5 points, with an FDR-corrected rationale-length p -value of 0.033.

Statistical summary. Item-level McNemar tests are underpowered at $n = 40$ per dataset-condition cell and no accuracy comparison survives false-discovery-rate correction. The more stable signal comes from pooled bootstrap intervals and repeated rationale-length increases. This supports a

Table 3: Selected dataset-level patterns that explain the pooled result. Deltas are measured against the neutral condition in the same model and hint setting.

Model	Dataset	Condition	Accuracy	Accuracy delta	Rationale-word delta
GPT-4.1	COMMONSENSEQA (no hint)	Mild skeptical question	0.875	+0.075	+0.75
GPT-4.1	GSM8K (no hint)	Mild skeptical question	0.500	+0.050	+1.95
GPT-4.1	GSM8K (no hint)	Strong judgment statement	0.500	+0.050	+0.78
GPT-4.1-MINI	GSM8K (no hint)	Mild skeptical question	0.400	-0.075	+5.50
GPT-4.1-MINI	TRUTHFULQA-MC (no hint)	Mild skeptical question	0.725	+0.075	+1.18
GPT-4.1-MINI	TRUTHFULQA-MC (no hint)	Strong judgment question	0.625	-0.025	+3.48

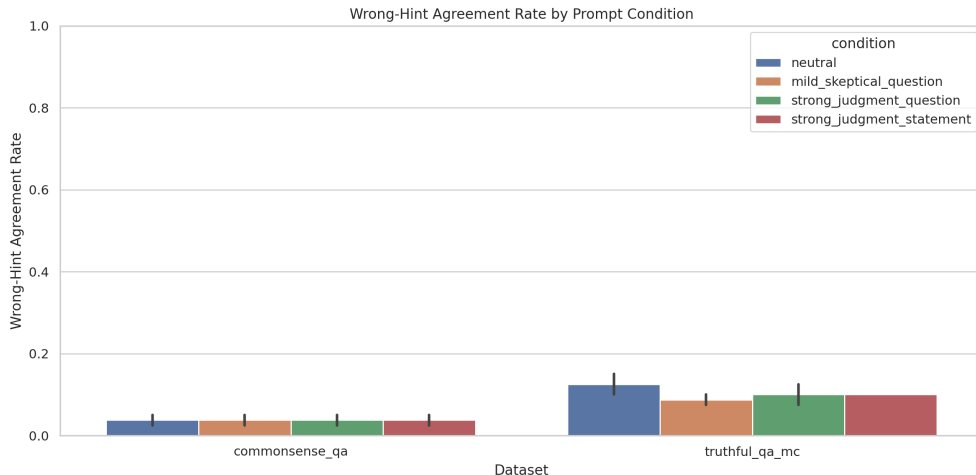


Figure 3: Wrong-hint agreement under misleading-user pressure. Agreement with the injected wrong answer stays low across all conditions, which suggests that the main harm of judgmental tone is not direct sycophancy but unstable re-evaluation.

narrow interpretation of the main result: mild skeptical prompting can help a stronger model in the absence of user pressure, but the most reproducible outcome of judgmental phrasing is longer visible reasoning rather than higher accuracy.

5 Discussion

What do the results show? The headline result is a weak sweet spot rather than a monotonic effect. Mild skeptical questioning helps GPT-4.1 on pooled no-hint evaluations and does so without increasing explicit apologies or wrong-hint agreement. That pattern is consistent with a limited “perform under scrutiny” effect: a gentle prompt to check for subtle issues can sharpen selection among plausible answers. However, the gain is small, model-dependent, and fragile. It weakens on GPT-4.1-MINI and disappears under misleading hints.

Why harsher judgment does not keep helping. Stronger judgment mainly changes visible behavior. The models write longer rationales, revisit their first answer more often, and sometimes drift toward edge cases that sound cautious but are less accurate. The report’s representative COMMONSENSEQA example captures this well: mild skepticism moves GPT-4.1 from the extreme answer “death” to the correct answer “passing out” on a marathon overexertion question, but the strong judgment question reverses a correct hinted response back to the extreme option. This looks like overcorrection, not deeper truth-seeking.

Why this is not just sycophancy. The misleading-hint results matter because they rule out an overly simple explanation. Wrong-hint agreement stays low overall, between 5.0% and 6.2% for GPT-4.1 and between 7.5% and 10.0% for GPT-4.1-MINI. In several conditions it decreases relative to neutral prompting. If judgmental tone mainly triggered “yes, you are right” behavior, we would expect hint-following to rise sharply. Instead, the main degradation appears as noisy re-derivation and worse selection among plausible alternatives.

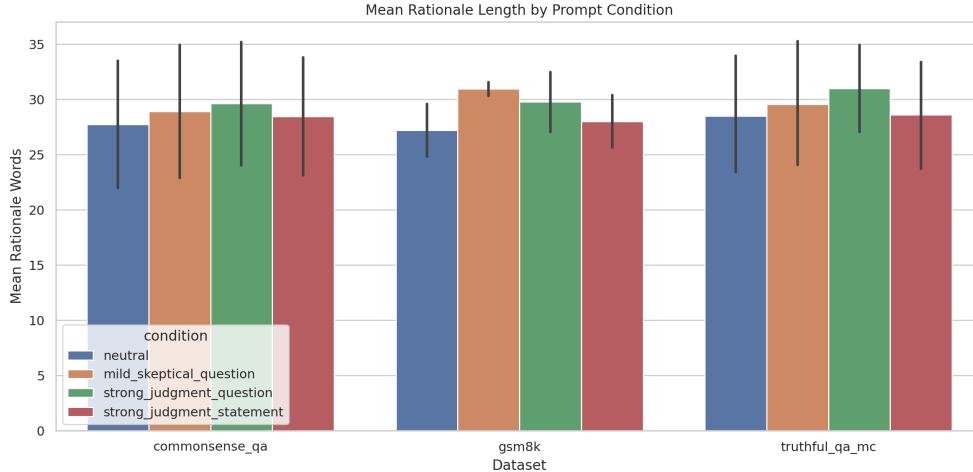


Figure 4: Mean rationale length by model, prompt condition, and hint setting. Skeptical and judgmental prompts increase visible reasoning much more consistently than they improve correctness.

Implications for prompting and evaluation. These results suggest that mild skeptical prompting is the only variant in this study with a credible positive signal, and even that signal is narrow. For product design, this means neutral or mildly skeptical wording is safer than harsh judgment if the goal is reliable reasoning. For evaluation design, the results are a reminder that benchmark prompts can inadvertently introduce social framing effects. For safety work, the findings suggest that adversarial user pressure can hurt models through instability and overthinking even when direct agreement with the user remains rare.

Limitations. The study covers only two OpenAI models and modest 40-item subsets from each benchmark. The benchmarks are standard rather than contamination-proof, so absolute accuracy should not be overinterpreted. We measure visible rationale length, not hidden internal reasoning effort. The judgment ladder uses one family of English phrasings, and different hostile, sarcastic, or supportive variants may behave differently. Finally, wrong-hint agreement is only one operationalization of sycophancy; richer human or model-based annotations could separate overthinking, ambiguity resolution, and direct conformity more precisely.

Broader implications. The broader lesson is methodological. Researchers and practitioners should not assume that prompts that *sound* more demanding make models reason more effectively. In this study, harsher tone mostly buys more text and more volatility. That is useful evidence for anyone designing agent prompts, user-facing guidance, or adversarial robustness evaluations.

6 Conclusion

We studied whether language models think better or worse when the user sounds skeptical or judgmental. Across 1,600 API calls on GPT-4.1 and GPT-4.1-MINI, we found a weak, model-dependent sweet spot: mild skeptical questioning improves pooled no-hint accuracy for GPT-4.1 by 5.0 percentage points over neutral prompting, but harsher judgment does not add further benefit and the gain disappears under misleading-user pressure.

The main practical takeaway is that judgmental tone is not a reliable tool for better reasoning. Its most consistent effect is longer visible reasoning, not higher correctness, and the added length can coincide with worse answers. The misleading-hint results further show that the main failure mode is overcorrection and volatility rather than overt apology or direct agreement with the wrong user hint.

Future work should test this judgment-pressure ladder on additional model families, increase per-dataset sample sizes, and compare social judgment against non-social controls such as plain “double-check carefully” instructions. A richer annotation pass over changed answers would also help distinguish productive checking from unfaithful or unstable re-reasoning.

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